Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If the tangent and normal to the hyperbola $x^2 - y^2 = 4$ at a point cut off intercepts a_1 and a_2 respectively on the x-axis, and b_1 and b_2 respectively on the y-axis, then the value of $a_1a_2 + b_1b_2$ is

(A) -1

(B) 0

(C) 4

(D) 1

Q.2

The normal at $P\left(x_1, y_1\right)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the coordinate axes at A and B. If O, is the origin and e, is the eccentricity of the hyperbola, then

(A) $OA = e^2x_1$

(B) $OB = e^2y_1$

(C) $OA = e^2y_1$

(D) $OB = e^2x_1$

Q.3

If the normal to the rectangular hyperbola $xy = c^2$ at the point $(ct, c/t)$ meets the curve again at $(ct', c/t')$ then

(A) $t^3t' = 1$

(B) $t^3t' = -1$

(C) $tt' = 1$

(D) $tt' = -1$

Q.4

Equations of the asymptotes of the hyperbola $3x^2 + 10xy + 8y^2 + 14x + 22y + 7 = 0$ is

(A) $3x^2 + 10xy + 8y^2 + 14x + 22y + 11 = 0$

(B) $3x^2 + 10xy + 8y^2 - 14x - 22y + 13 = 0$

(C) $3x^2 - 10xy + 8y^2 + 14x + 22y + 15 = 0$

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(D) $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$

Q.5

If the angle between the asymptotes of the hyperbola $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right)$ is 2θ and e is its eccentricity then $\sin\theta$ is equal to

(A) $\frac{b}{a}$

(B) $\frac{1}{e}$

(C) $\frac{a}{be}$

(D) $\sqrt{1 - \frac{1}{e^2}}$

Q.6

If the asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{36} = 1$ are perpendicular to the asymptotes of the hyperbola $\frac{x^2}{49} - \frac{y^2}{b^2} = 1$ then

(A) $7a \pm 6b = 0$

(B) $6a \pm 7b = 0$

(C) $a^2 - b^2 = 1$

(D) $a - b = 1$

Q.7

If θ is an angle between the two asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $\cos\left(\frac{\theta}{2}\right)$ is equal to

(A) $\frac{b}{\sqrt{a^2+b^2}}$

(B) $\frac{ab}{\sqrt{a^2+b^2}}$

(C) $\frac{a}{\sqrt{a^2+b^2}}$

(D) $\sqrt{\frac{a-b}{a+b}}$

Q.8

If the chords of contacts of the tangents from the points (x_1, y_1) and (x_2, y_2) to the hyperbola $2x^2 - 3y^2 = 6$ are at right angle, then $4x_1x_2 + 9y_1y_2$ is equal to

(A) -1

(B) 0

(C) 6

(D) -12

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Q.9

If $(a\sec\alpha, b\tan\alpha)$ and $(a\sec\beta, b\tan\beta)$ be the ends of a chord of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, passing through the focus $(ae, 0)$ then $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}$ is equal to:

(A) $\frac{1+e}{1-e}$

(B) $\frac{e+1}{e-1}$

(C) $\frac{1-e}{1+e}$

(D) $\frac{e-1}{e+1}$

Q.10

Let $P(\text{ a sec }\theta, b\tan\theta)$ and $Q(a\sec\phi, b\tan\phi)$,
where $\theta+\phi=\pi/2$, be two points on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If (h,k) is the point of intersection of the normals at P and Q, then k is equal to

(A) $\frac{a^2+b^2}{a}$

(B) $-\left(\frac{a^2+b^2}{a}\right)$

(C) $\frac{a^2+b^2}{b}$

(D) $-\left(\frac{a^2+b^2}{b}\right)$

Q.11

The value of m, for which the line
$y=mx+\frac{25\sqrt{3}}{3}$ is a normal to the conic
$\frac{x^2}{16}-\frac{y^2}{9}=1$, is

(A) $-\frac{2}{\sqrt{3}}$

(B) $\sqrt{3}$

(C) $-\frac{\sqrt{3}}{2}$

(D) None of these

Q.12

The equation of the normal to the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ at the point $(8, 3\sqrt{3})$ is

(A) $\sqrt{3}x+2y=25$

(B) $x+y=25$

(C) $y+2x=25$

(D) $2x+\sqrt{3}y=25$

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Q.13

The combined equation of the asymptotes of the hyperbola $2x^2 + 5xy + 2y^2 + 4x + 5y = 0$ is $-$

(A) $2x^2 + 5xy + 2y^2 + 4x + 5y + 2 = 0$

(B) $2x^2 + 5xy + 2y^2 + 4x + 5y - 2 = 0$

(C) $2x^2 + 5xy + 2y^2 = 0$

(D) None of these

Q.14

If $x = 9$ is the chord of contact of the hyperbola $x^2 - y^2 = 9$, then the equation of the corresponding pair of tangents is

(A) $9x^2 - 8y^2 + 18x - 9 = 0$

(B) $9x^2 - 8y^2 - 18x + 9 = 0$

(C) $9x^2 - 8y^2 - 18x - 9 = 0$

(D) $9x^2 - 8y^2 + 18x + 9 = 0$

Q.15

The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through the point $(3\sqrt{5}, 1)$ and the length of its latus rectum is $\frac{4}{3}$ units. The length of the conjugate axis is

(A) 2 units

(B) 3 units

(C) 4 units

(D) 5 units

Answers & Solutions

Q.1 Answer:

0

Solution:

Let the coordinates of a point P on the hyperbola be (x_1, y_1).

Equation of the tangent at P to the hyperbola is

$xx_1 - yy_1 = 4 \Rightarrow a_1 = \frac{4}{x_1}, b_1 = -\frac{4}{y_1}.$

Equation of the normal is

$y - y_1 = -\frac{y_1}{x_1}(x - x_1)$

$y_1x + x_1y = 2x_1y_1$

$a_2 = 2x_1, b_2 = 2y_1$

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So that

$a_1a_2 + b_1b_2 = 0$

Q.2 Answer:

$OA = e^2x_1$

Solution:

Equation of the normal at $P(x_1, y_1)$ **to the hyperbola** $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **is**

$\frac{x-x_1}{\frac{x_1}{a^2}} + \frac{y-y_1}{\frac{y_1}{b^2}} = 0$

$\Rightarrow OA = b^2\left(\frac{x_1}{a^2}\right) + x_1 = \frac{a^2+b^2}{a^2}x_1 = e^2x_1$

$OB = a^2\left(\frac{y_1}{b^2}\right) + y_1 = \frac{a^2+b^2}{b^2}y_1 = \frac{a^2e^2}{b^2}y_1$

Q.3 Answer:

$t^3t' = -1$

Solution:

Equation of the tangent at $(ct, c/t)$ **to the hyperbola** $xy = c^2$ **is**

$\frac{cx}{t} + cty = 2c^2$

Slope of the tangent $= -1/t^2$ **and slope of the normal** $= t^2$ **Equation of the normal at** $(ct, c/t)$ **is**

$y - c/t = t^2(x - ct)$ **If it passes through** $(ct', c/t')$**, then**

$\left(\frac{c}{t'} - \frac{c}{t}\right) = t^2(ct' - ct)$

$\Rightarrow (t - t') = t^3t'(t' - t) \Rightarrow t^3t' = -1$

Q.4 Answer:

$3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$

Solution:

Since the equation of a hyperbola differs from the combined equation of the asymptotes by a constant. Let the equation of the asymptotes be. $3x^2 + 10xy + 8y^2 + 14x + 22y + \lambda = 0$

This will represent a pair of straight lines if $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

where $a = 3, b = 8, c = \lambda, f = 11, g = 7, h = 5$

$\Rightarrow \quad 3 \times 8 \times \lambda + 2 \times 11 \times 7 \times 5 - 3 \times 121 - 8 \times 49$

$-\lambda \times 25 = 0$

$\Rightarrow \quad \lambda = 15.$

Q.5 Answer:

$\sqrt{1 - \frac{1}{e^2}}$

Solution:

Equation of the asymptotes is $\frac{x}{a} \pm \frac{y}{b} = 0.$

So $\tan 2\theta = \frac{\frac{b}{a} - \left(-\frac{b}{a}\right)}{1 + \left(\frac{b}{a}\right)\left(-\frac{b}{a}\right)}$

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$\Rightarrow 2\theta = \tan^{-1}\frac{2(b/a)}{1-(b/a)^2} = 2\tan^{-1}\left(\frac{b}{a}\right)$

$\Rightarrow \theta = \tan^{-1}\frac{b}{a} \Rightarrow \tan\theta = \frac{b}{a}.$

$\Rightarrow \sin\theta = \frac{b}{a} \times \frac{1}{\sqrt{1+\frac{b^2}{a^2}}} = \frac{b}{ae} = \sqrt{\frac{a^2(e^2-1)}{a^2e^2}}$

$= \sqrt{1-\frac{1}{e^2}}$

Q.6 Answer:

$7a \pm 6b = 0$

Solution:

Asymptotes to the two hyperbola are

$\frac{x}{a} \pm \frac{y}{6} = 0$ **and** $\frac{x}{7} \pm \frac{y}{b} = 0$

If they are at right angles then $\left(\pm\frac{6}{a}\right)\left(\pm\frac{b}{7}\right) = -1$

$\Rightarrow 7a \pm 6b = 0$

Q.7 Answer:

$\frac{a}{\sqrt{a^2+b^2}}$

Solution:

Slopes of the asymptotes are $\pm\frac{b}{a}$

$\Rightarrow \tan\theta = \frac{\frac{b}{a}-\left(-\frac{b}{a}\right)}{1+\left(\frac{b}{a}\right)\left(-\frac{b}{a}\right)}$

$\Rightarrow \frac{2\tan\left(\frac{\theta}{2}\right)}{1-\tan^2\left(\frac{\theta}{2}\right)} = \frac{2(b/a)}{1-\left(\frac{b}{a}\right)^2}$

$\Rightarrow \tan\left(\frac{\theta}{2}\right) = \frac{b}{a}$ **as** $\tan\left(\frac{\theta}{2}\right) > 0$

$\Rightarrow \cos\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{1+\frac{b^2}{a^2}}} = \frac{a}{\sqrt{a^2+b^2}}$

Q.8 Answer:

0

Solution:

Equation of the chords of contacts are

$2xx_1 - 3yy_1 = 6, 2xx_2 - 3yy_2 = 6$

Since they are at right angles $\frac{2x_1}{3y_1} \times \frac{2x_2}{3y_2} = -1$

$\Rightarrow \quad 4x_1x_2 + 9y_1y_2 = 0$

Q.9 Answer:

$\frac{1-e}{1+e}$

Solution:

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Equation of the chord joining the given points is $\begin{vmatrix} x & y & 1 \\ a\sec\alpha & b\tan\alpha & 1 \\ a\sec\beta & b\tan\beta & 1 \end{vmatrix} = 0$

$bx(\tan\alpha - \tan\beta) - ay(\sec\alpha - \sec\beta) + ab(\sec\alpha\tan\beta - \sec\beta$

$\tan\alpha) = 0$

$\Rightarrow bx\sin(\alpha-\beta) - ay(\cos\beta - \cos\alpha) + ab(\sin\beta - \sin\alpha) = 0$

$\Rightarrow bx\cos\frac{\alpha-\beta}{2} - ay\sin\frac{\alpha+\beta}{2} - ab\cos\frac{\alpha+\beta}{2} = 0$

$\Rightarrow \frac{x}{a}\cos\frac{\alpha-\beta}{2} - \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha+\beta}{2}$

If it passes through $(ae, 0)\text{e}\cos\frac{\alpha-\beta}{2} = \cos\frac{\alpha+\beta}{2}$

$\Rightarrow \frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}} = e$

$\Rightarrow \frac{\cos\frac{\alpha+\beta}{2} - \cos\frac{\alpha-\beta}{2}}{\cos\frac{\alpha-\beta}{2} + \cos\frac{\alpha-\beta}{2}} = \frac{e-1}{e+1}$

$\Rightarrow -\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{e-1}{e+1}$

$\Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{1-e}{1+e}$

Q.10 Answer:

$-\left(\frac{a^2+b^2}{b}\right)$

Solution:

Equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **at the point** $(a\sec\alpha, b\tan\alpha)$ **is given by** $ax\cos\alpha + by\cot\alpha = a^2 + b^2$

Normals at θ, ϕ **are** $\begin{cases} ax\cos\theta + by\cot\theta = a^2 + b^2 \\ ax\cos\phi + by\cot\phi = a^2 + b^2 \end{cases}$

where $\phi = \frac{\pi}{2} - \theta$ **and these passes through** $(h, k) \therefore \quad ah\cos\theta + bk\cot\theta = a^2 + b^2$

$\Rightarrow \quad ah\sin\theta + bk\tan\theta = a^2 + b^2$

Eliminating $h, bk(\cot\theta\sin\theta - \tan\theta\cos\theta)$

$= \left(a^2 + b^2\right)(\sin\theta - \cos\theta)$ **or** $k = -\left(a^2 + b^2\right)/b$

Q.11 Answer:

$-\frac{2}{\sqrt{3}}$

Solution:

The equation of normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **in terms of slope 'm' is**

$y = mx \pm \frac{m\left(a^2+b^2\right)}{\sqrt{a^2-b^2m^2}}$; **Given line**

$y = mx + \frac{25\sqrt{3}}{3}$

and conic $\frac{x^2}{16} - \frac{y^2}{9} = 1$ **which is hyperbola with** $a^2 = 16, b^2 = 9$

By comparing given line with equation of normal we get $\pm\frac{m\left(a^2+b^2\right)}{\sqrt{a^2-b^2m^2}} = +\frac{25\sqrt{3}}{3}$

$\Rightarrow \frac{m(16+9)}{\sqrt{16-9m^2}} = -\frac{25\sqrt{3}}{3}$

$\Rightarrow \frac{25m}{\sqrt{16-9m^2}} = -\frac{25\sqrt{3}}{3}$

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$\Rightarrow 9m^2 = 3\left(16 - 9m^2\right) \Rightarrow m^2 = \frac{16}{12} = \frac{4}{3}$
$\Rightarrow m = \pm\frac{2}{\sqrt{3}}$

Q.12 Answer:

$2x + \sqrt{3}y = 25$

Solution:

Applying the formula, the required normal is $\frac{16x}{8} + \frac{9y}{3\sqrt{3}} = 16 + 9$ **i.e.**
$2x + \sqrt{3}y = 25$ **Trick : This is the only equation among the given options at which the point** $(8, 3\sqrt{3})$ **is located**

Q.13 Answer:

$2x^2 + 5xy + 2y^2 + 4x + 5y + 2 = 0$

Solution:

Let the equation of asymptotes be $2x^2 + 5xy + 2y^2 + 4x + 5y + \lambda = 0 \quad \ldots\ldots(1)$
This equation represents a pair of straight lines, therefore $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$
$4\lambda + 25 - \frac{25}{2} - 8 - \lambda \times \frac{25}{4} = 0$
$\Rightarrow -\frac{9\lambda}{4} + \frac{9}{2} = 0 \Rightarrow \lambda = 2$
Putting the value of λ **in eq. (1), we get** $2x^2 + 5xy + 2y^2 + 4x + 5y + 2 = 0$
this is the equation of the asymptotes.

Q.14 Answer:

$9x^2 - 8y^2 - 18x + 9 = 0$

Solution:

(b) The equation of tangent at a point (x_1, y_1) **on the hyperbola** $x^2 - y^2 = c$ **is given by** $xx_1 - yy_1 = c$
Chord $x = 9$ **meets** $x^2 - y^2 = 9$ **at** $(9, 6\sqrt{2})$ **and**
$(9, -6\sqrt{2})$ **at which tangents are**
$9x - 6\sqrt{2}y = 9$ **and** $9x + 6\sqrt{2}y = 9$
or $3x - 2\sqrt{2}y - 3 = 0$ **and** $3x + 2\sqrt{2}y - 3 = 0$
$\therefore$ **Combined equation of tangents is** $(3x - 2\sqrt{2}y - 3)(3x + 2\sqrt{2}y - 3) = 0$
or $9x^2 - 8y^2 - 18x + 9 = 0$

Q.15 Answer:

4 units

Solution:

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
Hyperbola passes through $(3\sqrt{5}, 1)$
$\therefore \quad \frac{(3\sqrt{5})^2}{a^2} - \frac{1}{b^2} = 1$

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$\frac{45}{a^2}-\frac{1}{b^2}=1$ **...(i)**

Now length of latus rectum $=\frac{2\,\mathrm{b}^2}{\mathrm{a}} \Rightarrow \frac{4}{3}=\frac{2b^2}{a}$

$\Rightarrow \frac{2}{3}=\frac{b^2}{a} \Rightarrow a=\frac{3b^2}{2}$ **...(ii)**

Putting the value of ' a **' from equation (ii) in equation (i),**

$\Rightarrow \frac{45\times 4}{9b^4}-\frac{1}{b^2}=1 \Rightarrow \frac{20}{b^4}-\frac{1}{b^2}=1$

$20-b^2=b^4$

$b^4+b^2-20=0$

$b^4+5b^2-4b^2-20=0$

$b^2\left(b^2+5\right)-4\left(b^2+5\right)=0$

$\left(b^2-4\right)\left(b^2+5\right)=0$

$b^2=4, b^2=-5$

$\therefore \quad \mathrm{b}^2=4 \Rightarrow \mathrm{b}=2$

Now length of conjugate axis $=2\,\mathrm{b}=2(2)=4$

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